

- 15** Glucose isomerase is an enzyme often used in commercial food production as it catalyses the conversion of glucose to fructose, which has a sweeter taste. The enzyme is immobilised onto a surface or within agarose beads as part of this process.

The reaction is carried out between 55 °C and 60 °C.

Which statements explain why carrying out the reaction at this temperature is beneficial?

- 1 Immobilised enzymes contain more active sites at higher temperatures.
- 2 The activation energy of the reaction is more likely to be reached.
- 3 Increased kinetic energy results in more collisions between enzyme and substrate.
- 4 Maintaining this temperature is cheaper than carrying it out at lower temperatures.

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4